Answers and talk notes

Area of Regular Polygons

What your student is learning

Tonight we find the area of a regular polygon by splitting it into identical triangles — and then use that area to price a real job.

Answers

Ask your student to explain before you look.

1. **B.** 8 × 6. 8 × 6 gives the area: 6 equal triangles of 8 each.
2. **A.** 40 m². 40 m²: 8 × 5 = 40.
3. **D.** 36 m². 36 m²: one triangle is ½ × 4 × 3 = 6, and 6 × 6 = 36.
4. **C.** $144. $144: 36 × 4 = 144.
5. **A.** $120. $120: 60 × 2 = 120.

Talk about it

Why do you multiply one triangle's area by the number of sides instead of adding?

Home mission

Pick a plate or a rug and imagine it as a hexagon of 6 triangles. If one triangle is 5 square units and tile costs $2 per unit, how would you find the cost out loud?

When they are stuck, ask

- How many identical triangles does this shape split into?
- Did you find one triangle, or the whole shape?
- What does the price tell us we multiply by?

Try not to say

- Add the number of sides to the triangle's area.
- One triangle's area is the answer.
- The cost per square foot is the total cost.